

Geometric Dark Energy and the Spin-Torsion Bounce: A Complete Theoretical Assessment

Houston Golden^{1,*}

¹*Independent Researcher, Los Angeles, California, USA*

(Dated: March 16, 2026)

We present a comprehensive theoretical assessment of the Einstein-Cartan-Holst (ECH) framework as both a dark-energy phenomenology and a bouncing cosmology. The paper has three parts. First, we describe a phenomenological dark-energy ansatz rooted in ECH gravity that reduces fine-tuning from 10^{120} to $\sim 10^5$ and yields MCMC fits partially reducing the H_0 and σ_8/S_8 tensions ($H_0 = 69.2 \pm 0.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\sigma_8 = 0.785 \pm 0.016$). Second, we report the systematic closure of four minimal and three extended first-principles routes to deriving the late-time vacuum term, spanning Foundations A through G (seven structural barriers). Third, we present a comprehensive characterization of the spin-torsion bounce’s observable prospects across all perturbation channels: the tensor power spectrum is $P_T \approx 2 \times 10^{-64}$ (flat, $n_T \approx 0$), the chirality is $\Delta\chi = 0$ exactly (the effective four-fermion interaction $(J^5)^2$ is parity-*even*), the scalar transfer function is $T(k) = 1$ for all observable modes (time-reversal symmetry), and bounce-triggered state selection is prevented by Liouville’s theorem. The Poincaré gauge theory (PGT) extension lowers the bounce scale but the mass-coupling lock ($g_{\text{eff}} \sim m_T/M_{\text{Pl}}^2$) suppresses torsion-specific features, and the vacuum amplification ceiling ($\Omega_{\text{GW}} \propto (H/M_{\text{Pl}})^2$) ensures a minimum 10^{17} gap to any gravitational-wave detector. Two further routes are tested and closed: baryogenesis and relic production (Branch N) fail because at $T \sim M_{\text{Pl}}$ the torsion $(J^5)^2$ interaction is one of ~ 100 gravitational-strength channels (gravitational democracy), and hidden-sector vacuum selection (Branch O) fails because the bounce determines *whether* a transition occurs but not *what* vacuum energy results (bounce–vacuum energy decoupling). Finally, the PGT Sector II pseudoscalar torsion mode—the last candidate for a torsion-specific relic—is protected by an exact $\mathbb{Z}_2$ parity symmetry that prevents population at the bounce (Branch P). Fourteen structural barriers are cataloged across ten research branches (H through P), spanning five failure modes. The spin-torsion bounce is theoretically consistent but observationally inert: what is detectable is generic to any bounce; what is specific to torsion is undetectable.

PACS numbers: 98.80.-k, 04.50.Kd, 04.60.Pp, 95.36.+x

CONTENTS

I. Introduction	2	B. Branch G v1: strict one-loop	5
A. Relationship to prior work	3	C. Route T1: dynamical Immirzi field	5
II. Einstein-Cartan-Holst Framework	3	D. Route S1: parity CMB phenomenology	5
A. Action and field content	3	E. Foundations A–G: extended program	5
B. Torsion equation of motion	3	F. Closure summary	6
C. Reduced action after torsion elimination	3	VI. Structural Lessons	6
D. Modified Friedmann equation	4	A. Lesson 1: Algebraic torsion washes out	6
III. Phenomenological Dark-Energy Ansatz	4	B. Lesson 2: Fixed γ leaves no IR fingerprint	6
A. The scaling argument	4	C. Lesson 3: Dynamical γ reduces to generic ALP	6
B. Cosmological parameters	4	D. Lesson 4: Parity alone is not a mechanism	6
1. MCMC results	4	E. Lesson 5: No scale protection in the minimal model	6
2. Independent verification	4	F. Lesson 6: The mass-coupling lock	6
C. Fine-tuning reduction	4	VII. Bounce Tensor Spectrum	6
IV. Dark Energy Derivation Program	5	A. Background solution	6
A. Four minimal routes tested	5	B. Tensor perturbation equation	7
V. Closure Results	5	C. Bogoliubov coefficients	7
A. Track B: condensate route	5	D. Tensor power spectrum	7
		E. Chirality analysis	7
		F. Characteristic frequency	7
		VIII. Scalar Perturbations	8
		A. Bardeen equation through the bounce	8

* houston@hubify.com

B. Transfer function	8	D. What would change the assessment	14
C. Non-Gaussianity	8	E. The program as a publishable negative result	14
D. Pre-bounce dependence	8		
E. Summary: scalar sector	8	XVIII. Conclusion	14
IX. Bounce-Compatible Dark Energy and State Selection	8	Acknowledgments	15
A. Horndeski stability at the bounce (Branch I)	8	A. Notation and Conventions	15
B. State selection (Branch J)	8	B. Parameter Summary	15
X. UV→IR Bridge Extensions	9	C. Claims Classification	15
A. Screening results	9	D. Bardeen Equation Through the Bounce	15
B. PGT bounce parameters	9	1. Setup	15
XI. PGT Ghost-Free Parameter Space	9	2. Derivation of the Bardeen equation	16
A. Sector II: spin-0 ⁻ axial torsion	9	3. Time-reversal symmetry and $T(k) = 1$	16
B. Sector I: rejected	9	E. PGT Ghost-Free Conditions	16
C. Mass-coupling lock reappears	9	1. Quadratic PGT action	16
XII. PGT Bounce Gravitational Wave Spectrum	10	2. Single-mode ghost-free models	16
A. Background evolution	10	3. Sector II analysis	17
B. Effective potential and mode equation	10	4. Symmetry analysis	17
C. Bogoliubov spectrum	10	References	18
D. GW energy density: the critical result	10		
E. Amplitude-frequency tradeoff	10		
F. Minimum detector gap	10		
G. Comparison with other GW sources	10		
H. Barrier 12: Vacuum amplification ceiling	11		
XIII. Baryogenesis and Relic Production	11		
A. Candidate mechanisms screened	11		
B. Results	11		
C. Barrier 13a: Gravitational democracy	11		
XIV. Hidden-Sector Vacuum Selection	11		
A. Candidate mechanisms screened	11		
B. Results	12		
C. Barrier 13b: Bounce–vacuum energy decoupling	12		
D. Completeness of the state-change closure	12		
XV. PGT Torsion Relic Cosmology	12		
A. FRW parity protection	12		
B. Population mechanisms tested	12		
C. Literature context	13		
D. Barrier 14: Parity protection of the torsion vacuum	13		
XVI. Complete Barrier Catalog	13		
XVII. Discussion	13		
A. The generic-specific dichotomy	13		
B. Comparison with other bouncing cosmologies	13		
1. Loop quantum cosmology (LQC)	13		
2. Ekpyrotic bounce	13		
3. Matter bounce	13		
C. The complementary trap	13		

I. INTRODUCTION

The cosmological constant problem—why the observed vacuum energy density $\rho_\Lambda \sim (2.3 \text{ meV})^4$ is $\sim 10^{120}$ times smaller than the naive quantum field theory estimate $\sim M_{\text{Pl}}^4$ [1]—remains one of the deepest unsolved problems in fundamental physics. Growing tensions within Λ CDM, including the $\sim 4.9\sigma$ Hubble tension [2, 3] and $2\text{--}3\sigma$ σ_8/S_8 tension [4, 5], together with hints of dynamical dark energy from DESI DR2 BAO measurements [6, 7], motivate exploration of geometric approaches to dark energy.

Einstein-Cartan-Holst (ECH) gravity—Einstein-Cartan theory supplemented by the Holst term with Barbero-Immirzi parameter γ —is a natural starting point for geometric dark energy. The theory is well-motivated by loop quantum gravity [8], introduces spin-torsion coupling with parity-violating structure [9, 10], and provides a non-singular quantum bounce [12]. The central questions are whether this theoretical structure can (i) generate a natural origin for late-time dark energy, and (ii) produce distinctive observable signatures from the bounce.

This paper reports three sets of results, organized as a three-part assessment.

Part I: Framework and Phenomenology (Secs. II–III). We present the ECH framework and a phenomenological dark-energy ansatz that is observationally consistent and reduces fine-tuning significantly. MCMC fits partially reduce the H_0 and σ_8 tensions. The equation of state $w = -1$ is assumed, not derived.

Part II: Dark Energy Derivation Program (Secs. IV–VI). We report the systematic failure of four minimal first-principles routes to derive the late-time $w = -1$ behavior, together with the assessment of three extended foundations (A through G), identifying seven structural barriers and six structural lessons.

Part III: Bounce Observable Signatures (Secs. VII–XII). We present a comprehensive characterization of the spin-torsion bounce across all observable channels: tensor perturbations (Branch H), scalar perturbations (Branch K), bounce-compatible dark energy (Branch I), state selection (Branch J), UV→IR bridge extensions (Branch L), and the PGT gravitational wave spectrum (Branch M). Five additional barriers are identified, bringing the total to twelve.

The combination establishes the current scientific status of the program:

- The ECH framework is a *motivated phenomenological model*, not a first-principles derivation of dark energy.
- All tested routes to deriving $w = -1$ from geometry have failed, for distinct structural reasons spanning seven barriers.
- The spin-torsion bounce is observationally inert in the minimal model. The PGT extension produces a well-defined but undetectable gravitational wave spectrum.
- Twelve structural barriers, spanning three failure modes (energy-budget, signature, state-selection), prevent any nontrivial connection between the bounce and dark energy or between the bounce and cosmological observations.

A. Relationship to prior work

This paper supersedes an earlier version [20] (Paper 1) that presented the phenomenological framework organized around the minimal ECH model as if it were a sufficient foundation. The present version incorporates the results of a structured derivation and bounce characterization program (Branches H through M) that tested all natural routes and found all to be negative or generic. The phenomenological results are unchanged; the framing, scope, and conclusions are substantially extended to include the bounce sector.

II. EINSTEIN-CARTAN-HOLST FRAMEWORK

A. Action and field content

The starting point is the Einstein-Cartan-Holst action minimally coupled to N_f Dirac fermions:

$$S_{\text{ECH}} = \frac{M_{\text{Pl}}^2}{2} \int [\varepsilon_{IJKL} + \frac{2}{\gamma} \eta_{I[K} \eta_{L]J}] e^I \wedge e^J \wedge F^{KL}[\omega] + \int d^4x |e| [\bar{\psi} i \gamma^\mu D_\mu \psi - m \bar{\psi} \psi], \quad (1)$$

with $D_\mu = \partial_\mu + \frac{1}{4} \omega_\mu^{IJ} \gamma_I \gamma_J$ the Lorentz-covariant derivative and $\gamma = 0.274 \pm 0.020$ the Barbero-Immirzi parameter fixed by loop quantum gravity black hole entropy calculations [8]. The first term is the Holst action: the Palatini formulation of general relativity supplemented by the parity-odd Holst term $(1/\gamma) \eta_{I[K} \eta_{L]J} e^I e^J F^{KL}$. The second term is the standard minimal Dirac coupling.

B. Torsion equation of motion

In ECH gravity, the spin connection ω_μ^{IJ} is an independent dynamical variable, and its equation of motion is algebraic—torsion is *not propagating*. Solving the torsion equation exactly decomposes the connection as $\omega = \hat{\omega} + K$, where $\hat{\omega}$ is the torsion-free Levi-Civita connection and K is the contorsion tensor determined algebraically by the fermion spin density [9, 14].

C. Reduced action after torsion elimination

Substituting the algebraic torsion solution back into the action yields the reduced action [9, 13]:

$$S_{\text{red}} = S_{\text{EH}}[g] + S_{\text{Dirac}}[g, \psi] + S_{4f}[\psi; \gamma], \quad (2)$$

where S_{EH} is the standard Einstein-Hilbert action, S_{Dirac} is the standard minimally-coupled Dirac action with the Levi-Civita connection, and S_{4f} is the torsion-induced four-fermion contact interaction:

$$\mathcal{L}_{4f} = -G_{\text{eff}} \left(\bar{\psi} \gamma^\mu \gamma^5 \psi + \frac{1}{\gamma} \bar{\psi} \gamma^\mu \psi \right)^2, \quad G_{\text{eff}} = \frac{3\kappa^2}{16} \frac{\gamma^2}{\gamma^2 + 1}. \quad (3)$$

This perfect-square structure constrains the Fierz channel decomposition. At $\gamma = 0.274$, the vector admixture $1/\gamma \approx 3.65$ dominates; the scalar/pseudoscalar channel coupling $G_{\text{SP}} \propto (\gamma^2 - 1)/(\gamma^2 + 1)$ is repulsive (negative).

a. Critical structural observation. After torsion elimination, *all* γ -dependence resides in the four-fermion term S_{4f} . The Dirac kinetic operator $i\gamma^\mu \hat{\nabla}_\mu$ is the standard Levi-Civita operator, independent of γ . This structural fact is the root cause of the failures reported in Sec. V.

D. Modified Friedmann equation

The spin-torsion four-fermion interaction generates a ρ^2 correction to the Friedmann equation at high densities. When the fermion content is treated as a perfect fluid with energy density ρ , the effective Friedmann equation reads [12]:

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_{\text{crit}}}\right), \quad (4)$$

with critical density

$$\rho_{\text{crit}} \simeq 0.21 M_{\text{Pl}}^4, \quad (5)$$

where the numerical prefactor depends on the number of fermion species and γ . Equation (4) implies a non-singular bounce: $H = 0$ when $\rho = \rho_{\text{crit}}$, and $\dot{a} > 0$ for $\rho < \rho_{\text{crit}}$. The bounce replaces the classical Big Bang singularity with a smooth transition from contraction to expansion.

III. PHENOMENOLOGICAL DARK-ENERGY ANSATZ

A. The scaling argument

The ECH framework contains a parity-odd vacuum operator $\varepsilon^{abcd} K_{ab} R_{cd}$ sourced by the fermion spin density via torsion. At early times (Planck-density bounce), this operator has a natural magnitude set by $\alpha/M \sim \mathcal{O}(\kappa^2)$, where α/M is the parity-odd coefficient in the effective Lagrangian.

Inflationary expansion after the bounce dilutes this operator by a factor $\mathcal{D}_{\text{inf}} \sim e^{-3N_{\text{tot}}}$, where $N_{\text{tot}} \sim 92$ e -folds. The resulting late-time dark-energy density is parametrically

$$\rho_{\Lambda} \approx \Xi M_{\text{Pl}}^4, \quad \Xi \equiv [(\alpha/M) M_{\text{Pl}}] \mathcal{D}_{\text{inf}}, \quad (6)$$

giving $\Xi \sim 10^{-122}$ for appropriate values of α/M and N_{tot} .

a. What this is. Equation (6) is a *phenomenological scaling ansatz*. It traces a dimensional chain from the Planck scale through inflationary dilution to the observed vacuum energy. It reduces fine-tuning from 10^{120} (the cosmological constant problem in Λ CDM) to $\sim 10^5$ (sensitivity to N_{tot} within $\Delta N \approx 4$ e -folds).

b. What this is not. Equation (6) is *not* a derivation of $w = -1$. The parity-odd operator is sourced by the spin density, which vanishes at late times ($K_{ab} \rightarrow 0$). For the residual to persist as a true vacuum energy, one must show that integrating out the spin degrees of freedom generates an IR-constant term in the effective action. Four attempts to establish this have failed (Sec. V). We therefore treat $w = -1$ as a phenomenological assumption, consistent with observations ($|w + 1| \lesssim 10^{-2}$ from DESI+Planck).

TABLE I. Cosmological parameters from MCMC fits. Posterior means $\pm 1\sigma$. Original analysis uses the full tension dataset; verification uses Cobaya v3.6.1 with CamSpec.

Parameter	Original	Verification
H_0 [km/s/Mpc]	69.2 ± 0.8	67.68 ± 1.06
σ_8	0.785 ± 0.016	0.803 ± 0.008
S_8	0.80 ± 0.02	0.814 ± 0.008
ΔN_{eff}	—	-0.020 ± 0.169
Ω_m	0.310 ± 0.008	0.308 ± 0.005
n_s	0.9649 ± 0.0042	—
τ	0.054 ± 0.007	—
Total samples	—	176,840
$\hat{R} - 1$	< 0.01	0.001

B. Cosmological parameters

The phenomenological framework motivates a Λ CDM+ ΔN_{eff} extension: the bounce scenario qualitatively predicts additional relativistic species from particle production near the Planck density, but quantitative estimates are not yet available.

1. MCMC results

MCMC fits to the full tension dataset (Planck 2018 + DESI DR1 BAO + Pantheon+ + SH0ES H_0 prior + DES Y3 S_8) yield:

$$H_0 = 69.2 \pm 0.8 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (7)$$

$$\sigma_8 = 0.785 \pm 0.016, \quad (8)$$

$$S_8 = 0.80 \pm 0.02. \quad (9)$$

These partially reduce the H_0 tension (from 4.9σ to 2.9σ vs. SH0ES) and the S_8 tension (from $\sim 3\sigma$ to $\sim 1\sigma$ vs. weak lensing).

2. Independent verification

Independent verification using Cobaya v3.6.1 with Planck NPIPE CamSpec TTTEEE + lowl TT/EE + lensing finds ΔN_{eff} consistent with zero ($\Delta N_{\text{eff}} = -0.020 \pm 0.169$ for the full-tension dataset, $\Delta N_{\text{eff}} = +0.065 \pm 0.17$ for Planck+BAO+SN without H_0/S_8 priors; see Table I). The tension reduction in the original analysis is driven by the SH0ES prior, not by the ΔN_{eff} extension itself.

C. Fine-tuning reduction

A Monte Carlo sensitivity scan (10^5 samples) confirms that the scaling ansatz Eq. (6) reduces fine-tuning from 10^{120} to $\sim 10^5$: the viable range $N_{\text{tot}} \in [79, 95]$

produces ρ_Λ within two orders of magnitude of the observed value, with N_{tot} as the single controlling parameter ($|\rho_s| = 0.996$). The residual tuning is concentrated in a single initial condition with potential dynamical answers through the bounce-to-inflation transition.

IV. DARK ENERGY DERIVATION PROGRAM

The central question is whether late-time $w = -1$ can be derived from first principles within the minimal ECH+Dirac model or its natural extensions. We tested four minimal routes and assessed three extended foundations (A through G), each with predefined gates and kill criteria frozen before computation.

A. Four minimal routes tested

Track B (NJL condensate):

Test whether a pseudoscalar fermion condensate $\langle \bar{\psi} i \gamma^5 \psi \rangle$ forms via the Nambu-Jona-Lasinio mechanism from the torsion-induced four-fermion interaction and persists as late-time vacuum energy.

Branch G v1 (strict one-loop effective action):

Compute the one-loop gravitational effective action $\Gamma[e]$ after torsion elimination and fermion integration, and test for γ -dependent vacuum content.

Route T1 (dynamical Immirzi field):

Promote $\gamma \rightarrow \theta(x)$ to a dynamical pseudoscalar and assess whether the reduced action contains novel content not reproducible by adding a standard axion-like particle (ALP) to GR.

Route S1 (parity-sensitive CMB phenomenology):

Assess whether the framework's parity-odd operator maps to distinctive birefringence observables distinguishable from generic ALP predictions.

All four routes were closed with clean negative results. Additionally, Foundations A through G (an extended program testing propagating torsion, Nieh-Yan non-topological coupling, environmental mass, disformal coupling, scale separation, attractor sensitivity, and vacuum selection) were all closed, yielding seven barriers in total.

V. CLOSURE RESULTS

A. Track B: condensate route

Fierz rearrangement of the four-fermion interaction (3) decomposes it into standard NJL channels. The

scalar/pseudoscalar coupling is

$$G_{\text{SP}} = \frac{3\kappa^2}{16} \frac{\gamma^2 - 1}{\gamma^2 + 1}, \quad (10)$$

which is *repulsive* (negative) for $\gamma < 1$. At $\gamma = 0.274$, the NJL gap equation has no nontrivial solution. Even for $\gamma > 1$ (attractive sign), the coupling is $\sim 175\times$ below the NJL critical coupling. Track B closes at Gate 1: no condensate forms.

B. Branch G v1: strict one-loop

After torsion elimination, the Dirac operator is $\not{D} = i\gamma^\mu \nabla_\mu$ —the standard γ -independent Levi-Civita operator. The four-fermion term is quartic in ψ and does not contribute at one loop. The Seeley-DeWitt coefficients a_0, a_1, a_2 are the standard textbook values with no γ -dependence [15]. Branch G v1 closes: no novel Holst content at strict one-loop.

C. Route T1: dynamical Immirzi field

Promoting $\gamma \rightarrow \theta(x)$, torsion remains algebraic. After elimination, the reduced action is [11, 16]

$$S_{\text{red}} = S_{\text{EH}} + S_{\text{Dirac}} + \int \left[-\frac{1}{2} f(\theta) (\partial\theta)^2 + \mathcal{L}_{4f} \right], \quad (11)$$

where $\mathcal{L}_{4f}$ is θ -independent. Every term is reproducible by adding a standard ALP to GR. The Nieh-Yan 4-form is exact (total derivative) in 4D; its topological nature ensures the gravitational origin washes out [16]. Cosmological dynamics give $w = +1$ (stiff matter), the opposite of dark energy [17]. Route T1 closes.

D. Route S1: parity CMB phenomenology

The minimal ECH action has no photon-polarization coupling. The parity-odd operator is purely gravitational; the electromagnetic field strength does not appear in the Holst term or the reduced action. Any assumed coupling produces constant- β birefringence with two free parameters for one observable—identical to generic ALP phenomenology. Route S1 closes.

E. Foundations A–G: extended program

Beyond the four minimal routes, three extended foundations were assessed:

- *Foundation A* (PGT propagating torsion): The ghost-free 0^- axial torsion mode has mass $m_B = M_{\text{Pl}}/(4\sqrt{\pi|t_3|})$ and matter coupling $g_{\text{eff}} \sim 1/(M_{\text{Pl}}\sqrt{|t_3|})$. The *mass-coupling lock*

TABLE II. Tested routes and outcomes. Four minimal routes plus Foundation A (the most natural PGT extension).

Route	Target	Method	Failure
Track B	Condensate	Fierz + NJL	$G_{\text{SP}} < 0$ at $\gamma < 1$
G v1	$\Gamma[e]$	1-loop	$\mathbb{D}$ is γ -free
T1	$\theta(x)$	Lit. review	= standard ALP
S1	Birefr.	Assessment	No photon coupling
Found. A	PGT 0^-	Phase 1+2	Mass-coupling lock
Found. B	Nieh-Yan	MAG analysis	Top.-shift duality
Found. C	Environ.	Scalar-tensor	FRW universality
Found. D	Disformal	Connection	Planck suppression
Found. E	Integrals	Scale sep.	k_{CMB}/k_b
Found. F	Attractors	IC analysis	Sensitivity dilemma
Found. G	Vacuum sel.	Cyclic	Parameter immunity

(Barrier 1): m_B and g_{eff} decrease together, with $m_B/g_{\text{eff}} \sim M_{\text{Pl}}^2$ fixed. At meV-scale mass, the coupling is $10^{29} \times$ below gravitational strength. No symmetry within PGT protects m_B .

- *Foundations B–D*: The Nieh-Yan non-topological route (B), environmental mass (C), and disformal coupling (D) were all closed by distinct mechanisms: topological-shift duality, scalar-tensor universality, and Planck suppression, respectively.
- *Foundations E–G*: Scale separation, attractor-sensitivity dilemma, and parameter immunity closed the remaining directions.

F. Closure summary

VI. STRUCTURAL LESSONS

The closures reveal a hierarchy of structural obstructions. Lessons 1–5 emerge from the minimal ECH model; Lesson 6 from the PGT extension.

A. Lesson 1: Algebraic torsion washes out

In the minimal ECH model, torsion is non-propagating. Its equation of motion is algebraic: con-torsion K is determined pointwise by the fermion spin density. Torsion elimination is exact and complete—no torsion-sector information survives into the reduced action except through the four-fermion coupling constants.

B. Lesson 2: Fixed γ leaves no IR fingerprint

γ enters the reduced action only through G_{eff} and G_{SP} in the four-fermion interaction. These are UV-scale contact couplings. At one loop, the fermion determinant is γ -independent. At the condensate level, the coupling is

subcritical. γ sets the UV-scale spin-torsion structure but does not generate low-energy phenomena.

C. Lesson 3: Dynamical γ reduces to generic ALP

The Nieh-Yan 4-form $N_4 = d(e^I \wedge T_I)$ is exact in 4D. Any coupling of a pseudoscalar θ to N_4 reduces to boundary terms plus standard kinetic/coupling structures. The gravitational origin is topologically protected from leaving a local dynamical imprint.

D. Lesson 4: Parity alone is not a mechanism

Without a derived coupling to observable sectors, the ECH action’s parity-odd sector produces no testable prediction distinguishing the framework from generic alternatives. Parity is a structural property, not by itself a mechanism or signal source.

E. Lesson 5: No scale protection in the minimal model

None of the tested routes addresses naturalness. The condensate route produces the wrong sign. The one-loop route has standard renormalization. The dynamical Im-mirzi route requires $V(\theta_0) = \Lambda_{\text{obs}}$ by hand.

F. Lesson 6: The mass-coupling lock

Even when torsion propagates (PGT), the parameter that controls the torsion mass also controls the matter coupling:

$$m_B \propto \frac{1}{\sqrt{|t_3|}}, \quad g_{\text{eff}} \propto \frac{1}{\sqrt{|t_3|}}, \quad \frac{m_B}{g_{\text{eff}}} \sim M_{\text{Pl}}^2. \quad (12)$$

Making the mode cosmologically light automatically makes it cosmologically inert. The lock occurs generically when a single parameter controls both the kinetic normalization and the mass of a geometric mode. No symmetry within PGT protects the mass, and graviton loops generate $\delta m_B^2 \sim M_{\text{Pl}}^2/(16\pi^2)$, requiring fine-tuning of 1 in 10^{57} for meV-scale mass.

VII. BOUNCE TENSOR SPECTRUM

A. Background solution

For a radiation-dominated fluid ($p = \rho/3$) on the modified Friedmann equation (4), the exact scale factor is

$$a(t) = a_b (1 + 4\alpha^2 t^2)^{1/4}, \quad (13)$$

where a_b is the minimum scale factor at the bounce ($t = 0$) and

$$\alpha = \sqrt{\frac{8\pi}{3}} M_{\text{Pl}} \quad (14)$$

is set by the Planck scale. At late times ($\alpha t \gg 1$), Eq. (13) reduces to $a \propto t^{1/2}$, the standard radiation-dominated expansion.

B. Tensor perturbation equation

Tensor perturbations h_{ij} on the bounce background satisfy the standard wave equation in conformal time η :

$$\mu_k'' + \left(k^2 - \frac{a''}{a}\right) \mu_k = 0, \quad (15)$$

where $\mu_k = a h_k$ is the Mukhanov variable, primes denote derivatives with respect to conformal time, and the effective potential is

$$\frac{a''}{a} = \frac{2\alpha^2 a_b^2}{(1 + 4\alpha^2 t^2)^{3/2}}. \quad (16)$$

The potential is *always positive*, localized within a time $\Delta t \sim 1/\alpha \sim t_{\text{Pl}}$ around the bounce, and defines a characteristic wavenumber $k_b = a_b \sqrt{2} \alpha \approx 1.88 a_b M_{\text{Pl}}$.

C. Bogoliubov coefficients

Numerical solution of Eq. (15) with vacuum initial conditions in the contracting phase yields the Bogoliubov coefficient:

$$|\beta_k|^2 \approx 0.855 \frac{k_b^2}{k^2} \quad \text{for } k \ll k_b, \quad (17)$$

with exponential suppression for $k \gg k_b$. The $1/k^2$ scaling is confirmed analytically from the Born approximation for scattering off the localized potential.

D. Tensor power spectrum

The tensor power spectrum is

$$P_T(k) \propto k^2 (1 + 2|\beta_k|^2) \approx \text{const} \quad \text{for } k \ll k_b. \quad (18)$$

The k^2 phase-space factor exactly cancels the $1/k^2$ from the Bogoliubov coefficient, producing a *flat* (scale-invariant) spectrum with

$$n_T \approx 0 \quad (\text{numerical: } n_T = -0.002 \pm 0.01). \quad (19)$$

The absolute amplitude, however, is suppressed by the enormous redshift from the bounce epoch to the present:

$$P_T \approx 2 \times 10^{-64}, \quad (20)$$

TABLE III. Parity-odd candidate mechanisms. None provides a tensor chirality source within the minimal model.

Candidate	Parity-odd?	Tensor coupling?
$(J^5)^2$ interaction	No (even)	—
n_5 background	Conditional	No
Holst/Immirzi	Rescales only	No
Dynamical Nieh-Yan	Yes (not EC)	Generic ALP
Chern-Simons $\tilde{R}R$	Yes (not EC)	Yes (not EC)
Chiral anomaly	Yes (quantum)	No ($\tilde{R}R = 0$ on FRW)

reflecting a dilution factor $(a_b/a_0)^2 \sim 10^{-65}$. The tensor-to-scalar ratio is correspondingly

$$r = \frac{P_T}{P_S} \sim \frac{10^{-64}}{10^{-9}} \sim 10^{-55}, \quad (21)$$

which is 51 orders of magnitude below any conceivable detection threshold.

E. Chirality analysis

A key question is whether the spin-torsion bounce produces chiral gravitational waves (different amplitudes for left- and right-circular polarizations). Six candidate parity-odd mechanisms were analyzed:

The result is unambiguous:

$$\Delta\chi(k) \equiv \frac{P_L - P_R}{P_L + P_R} = 0 \quad (\text{exact}). \quad (22)$$

The four-fermion interaction $(J_\mu^5)^2$ is the product of two pseudovectors, giving a *scalar*—it is parity-*even* despite arising from parity-odd torsion. FRW isotropy eliminates all spatial parity-odd backgrounds ($\langle J_i^5 \rangle = 0$), and the Pontryagin density vanishes identically on FRW ($\tilde{R}^{\mu\nu}{}_{\rho\sigma} R^{\rho\sigma}{}_{\mu\nu} = 0$). The bounce breaks time-reversal symmetry, not spatial parity.

This constitutes *Barrier 8: parity-even effective interaction*. The spin-torsion four-fermion interaction is parity-even; FRW isotropy eliminates all remaining parity-odd backgrounds; no chirality in tensor observables.

F. Characteristic frequency

The bounce tensor spectrum extends from $f = 0$ to a cutoff at

$$f_b \approx 8 \text{ GHz} \quad (23)$$

in the minimal model. This is far above any gravitational wave detector band.

VIII. SCALAR PERTURBATIONS

A. Bardeen equation through the bounce

Scalar perturbations of a radiation fluid on the bounce background are described by the Bardeen potential Φ , which satisfies

$$\ddot{\Phi} + 5H\dot{\Phi} + \left[\frac{k^2}{3a^2} + 2\dot{H} + 4H^2 \right] \Phi = 0. \quad (24)$$

Crucially, this equation is *regular* at $H = 0$ (the bounce instant), unlike the Mukhanov-Sasaki variable $v = z\mathcal{R}$ where $z \propto a\dot{\phi}/H$ diverges at the bounce. The Bardeen formulation avoids this singularity entirely, as noted by Peter and Pinto-Neto [29].

B. Transfer function

a. Analytic proof. The Bardeen equation possesses exact time-reversal symmetry on the spin-torsion bounce background. Under $\eta \rightarrow -\eta$, the growing mode of contraction maps to the decaying mode of expansion, and the constant mode maps to itself with unit amplitude. Therefore

$$T(k) = 1 \quad \text{for all } k \ll k_b. \quad (25)$$

The bounce does not modify the scalar power spectrum at any observable scale. Features appear only at $k \sim k_b$ (GHz frequencies).

b. Numerical confirmation. $T(k) = 1.000$ to machine precision for $k/k_b < 0.1$. The growing mode amplifies by $(k_b/k)^3 \sim 10^{84}$ for CMB modes during contraction, but the time-reversal symmetry maps this entirely to the decaying mode of the expanding phase. No matching ambiguity, no spectral distortion.

C. Non-Gaussianity

The torsion-specific $(J^5)^2$ interaction generates

$$f_{\text{NL}}^{\text{torsion}} \sim 10^{-56} \quad (26)$$

at CMB scales, suppressed by the $(k/k_b)^2$ localization of the bounce interaction. Generic mode coupling through gravity gives $f_{\text{NL}} \sim \mathcal{O}(1)$ at most—the same as any bounce. No distinctive non-Gaussian signal.

D. Pre-bounce dependence

The observed scalar spectrum (A_s , n_s , running) is entirely determined by the pre-bounce contraction mechanism, which the minimal model does not specify. A pure radiation contraction gives $P_S \propto k^4$ (excluded by CMB data). A viable spectrum requires a matter-dominated or ekpyrotic pre-bounce phase—physics beyond the minimal spin-torsion bounce.

TABLE IV. Horndeski model classes at the bounce.

Class	Verdict	Key finding
Quintessence	Trivial compat.	$(Q_S, c_S^2) = (1, 1)$
K-essence	Trivial compat.	$X \rightarrow 0$ at bounce
Braiding	EFT breaks	$Q_S \rightarrow 0$; $M \sim H_0 \ll M_{\text{Pl}}$
Non-minimal	Compatible	Tachyonic kick, bounded
Quartic/Quintic	Trivial compat.	Pre-empted by GW170817
DHOST	EFT breaks	Degeneracy algebraic

E. Summary: scalar sector

The scalar sector provides *no* distinctive bounce signature at observable scales. The scale separation $k_{\text{CMB}}/k_b \sim 10^{-28}$ ensures that the bounce is a sub-resolution event for all cosmological probes. Everything observable (the spectral shape, the transfer function) is generic to any time-symmetric radiation bounce; everything specific to spin-torsion (the bounce profile, the $(J^5)^2$ correction) is confined to GHz scales.

IX. BOUNCE-COMPATIBLE DARK ENERGY AND STATE SELECTION

A. Horndeski stability at the bounce (Branch I)

We tested whether the spin-torsion bounce ($H = 0$, $\dot{H} \approx 3.52 M_{\text{Pl}}^2$) constrains which Horndeski/scalar-tensor dark energy models are viable. Six model classes were screened for ghost freedom, gradient stability, and EFT validity at the bounce:

The root cause is scale separation: $\rho_{\text{DE}}/\rho_{\text{crit}} \sim 10^{-122}$. DE fields are frozen ($X \rightarrow 0$, $\dot{\phi} \sim 10^{-60} M_{\text{Pl}}^2$) during the Planck-scale bounce. All stability conditions reduce to their trivial limits.

A methodological finding: the EFT-of-DE α -parameterization (Bellini-Sawicki) is singular at $H = 0$ ($\alpha_K \propto 1/H^2$, $\alpha_B \propto 1/H$ diverge), while physical stability quantities remain finite. Standard Boltzmann codes cannot be applied to bouncing cosmologies without reformulation.

B. State selection (Branch J)

Five candidate mechanisms for bounce-triggered dark-energy state selection were tested: pNGB misalignment, multi-vacuum selection, symmetry re-breaking, metastable trapping, and nonadiabatic excitation. All five fail for three converging reasons:

a. Barrier 9: Hamiltonian phase-space conservation. The bounce is a Hamiltonian scattering event occurring near $H = 0$ (negligible friction). Liouville's theorem guarantees that the mapping from pre-bounce to post-bounce states preserves phase-space volume. The bounce *rotates* dark-sector initial conditions but cannot *contract*

them to a predictive outcome. Phase-space contraction requires dissipation or irreversibility, both absent at the bounce instant.

b. Naturalness dilemma. Any curvature coupling strong enough to affect the dark sector at the bounce ($\xi \sim \mathcal{O}(1)$) generates Planck-scale radiative corrections to the DE mass ($\delta m^2 \sim \xi M_{\text{Pl}}^2$), destroying the DE potential structure. A coupling weak enough to preserve the DE sector ($\xi \sim 10^{-124}$) is far too weak for state selection.

c. Scale separation. The 10^{61} hierarchy between M_{Pl} and m_{DE} prevents parametric resonance (frequency mismatch), meaningful field evolution during the bounce ($t_{\text{Pl}} \ll 1/m_{\text{DE}}$), and nonadiabatic effects at DE frequencies.

X. UV→IR BRIDGE EXTENSIONS

The scale separation $k_{\text{CMB}}/k_b \sim 10^{-28}$ is the meta-barrier underlying many of the null results above. Branch L screened seven candidate mechanisms for transferring bounce-scale physics to observable scales.

A. Screening results

- *PGT lower-scale bounce:* Survives. In Poincaré gauge theory, torsion propagates with mass m_{T} , and the effective bounce critical density becomes

$$\rho_{\text{crit}}^{\text{eff}} \sim m_{\text{T}}^2 M_{\text{Pl}}^2. \quad (27)$$

For $m_{\text{T}} \sim 10^3\text{--}10^9$ GeV, bounce features move to the LISA–ET frequency bands.

- *Bounce + brief inflation:* Killed—reduces to standard inflation with the bounce irrelevant.
- *Time-asymmetric matter bounce:* Killed—not bounce-specific (any bounce works).
- *Generic curvaton:* Killed—not bounce-specific.
- *Torsion-curvaton:* Conditional ($\sim 3\%$)—requires PGT with specific parameter tuning.
- *Bounce-triggered relics:* Killed—not bounce-specific.
- *Cyclic resonance:* Killed—pathological (Foundation G incompatibility).

The screening reveals a sharp dichotomy (*Barrier 10: UV→IR specificity dilemma*): mechanisms that beat scale separation generically (contraction-era spectrum generation, inflation) do not need the bounce, while the only mechanism that is bounce-specific (lowering ρ_{crit} via PGT) requires changing the gravitational theory itself.

TABLE V. PGT bounce parameter scan: torsion mass m_{T} vs. bounce frequency f_b and peak GW amplitude $\Omega_{\text{GW}}^{\text{peak}} h^2$. The detector gap is the ratio of detector sensitivity to signal amplitude.

m_{T} [GeV]	f_b [Hz]	$\Omega_{\text{GW}}^{\text{peak}} h^2$	Gap to best detector
10^{18} ($= M_{\text{Pl}}$)	2.6×10^{10}	5×10^{-6}	10^4 (HF)
10^{14}	2.6×10^8	5×10^{-14}	10^9 (ET)
10^{10}	2.6×10^6	5×10^{-22}	10^{12} (ET)
10^7	2.4×10^4	3×10^{-30}	10^{17} (ET)
10^4	2.6×10^2	5×10^{-34}	10^{21} (LIGO)
10^0	2.6	5×10^{-42}	10^{32} (DECIGO)
10^{-3}	8.2×10^{-2}	5×10^{-48}	10^{35} (LISA)

B. PGT bounce parameters

The PGT bounce frequency scales as

$$f_b \approx 2.6 \times 10^{10} \left(\frac{m_{\text{T}}}{M_{\text{Pl}}} \right)^{1/2} \text{ Hz}, \quad (28)$$

and the torsion mass determines the bounce scale:

$$m_{\text{T}} = \frac{M_{\text{Pl}}}{2\sqrt{|t_3|}}. \quad (29)$$

XI. PGT GHOST-FREE PARAMETER SPACE

A. Sector II: spin-0[−] axial torsion

In the quadratic PGT action, the ghost-free parameter subspace has been mapped by Sezgin and van Nieuwenhuizen [23] and Karananas [24]. The cleanest single-mode model is the spin-0[−] (pseudoscalar axial torsion), which is ghost-free for all $|t_3| > 0$ and has one light mode.

B. Sector I: rejected

The spin-0⁺ scalar torsion mode develops a ghost at the target mass scale required for observable bounce features. This sector is rejected.

C. Mass-coupling lock reappears

In the ghost-free 0[−] sector, canonical normalization gives

$$g_{\text{eff}} \sim \frac{m_{\text{T}}}{M_{\text{Pl}}^2}, \quad (30)$$

confirming the mass-coupling lock (Lesson 6): the effective matter coupling is $m_{\text{T}}/M_{\text{Pl}}^2$, which vanishes in the light-mass limit.

This constitutes *Barrier 11: decoupling universality*. The pattern is universal: any geometric mode whose kinetic term is inherited from the gravitational action without an independent mass-generating mechanism will exhibit this decoupling in the canonically normalized theory.

XII. PGT BOUNCE GRAVITATIONAL WAVE SPECTRUM

A. Background evolution

The PGT bounce background has the same functional form as the minimal model, Eq. (13), but with α replaced by

$$\alpha_{\text{PGT}} = m_{\text{T}} \sqrt{\frac{8\pi}{3}}, \quad (31)$$

reflecting the lowered bounce energy scale $\rho_{\text{crit}}^{\text{eff}} \sim m_{\text{T}}^2 M_{\text{Pl}}^2$.

B. Effective potential and mode equation

The tensor mode equation is identical to Eq. (15), with the effective potential

$$\frac{a''}{a} = \frac{2\alpha_{\text{PGT}}^2 a_b^2}{(1 + 4\alpha_{\text{PGT}}^2 t^2)^{3/2}}. \quad (32)$$

C. Bogoliubov spectrum

The Bogoliubov coefficient follows the same functional form as the minimal model:

$$|\beta_k|^2 \approx \begin{cases} \text{const} & k \ll k_b, \\ \exp(-\pi k/k_b) & k \gg k_b. \end{cases} \quad (33)$$

D. GW energy density: the critical result

The GW energy density spectrum is

$$\Omega_{\text{GW}}(f) h^2 \approx 5 \times 10^{-6} \left(\frac{m_{\text{T}}}{M_{\text{Pl}}} \right)^2 S(f/f_b), \quad (34)$$

where $S(x)$ is a universal shape function with $S \approx 1$ for $x \ll 1$, oscillatory with $1/x^2$ envelope near $x \sim 1$, and exponentially suppressed for $x \gg 1$.

The critical feature is the factor $(m_{\text{T}}/M_{\text{Pl}})^2$ —this is *not* $\mathcal{O}(1)$ as might be naively expected. The efficiency of vacuum amplification is

$$\varepsilon = \frac{\rho_{\text{GW}}}{\rho_{\text{crit}}} \sim \frac{m_{\text{T}}^2}{M_{\text{Pl}}^2} = \left(\frac{m_{\text{T}}}{M_{\text{Pl}}} \right)^2, \quad (35)$$

TABLE VI. Detector sensitivity comparison for the PGT bounce GW spectrum. The minimum gap across all frequencies and parameter choices is 10^{17} .

Detector	Band [Hz]	$\Omega_{\text{GW}}^{\text{sens}} h^2$	Min. gap
PTA	10^{-9} – 10^{-7}	10^{-10}	10^{42}
LISA	10^{-4} – 10^{-1}	10^{-13}	10^{35}
DECIGO	10^{-1} – 10^1	10^{-16}	10^{26}
LIGO O5	10^1 – 10^3	10^{-10}	10^{20}
ET	10^0 – 10^4	10^{-13}	10^{17}
HF concept	10^8 – 10^{10}	10^{-10}	10^4

which is the standard graviton production efficiency: $\mathcal{O}(1)$ gravitons per mode, but each graviton carries energy $\sim m_{\text{T}}$ while the total energy budget is $\sim m_{\text{T}} M_{\text{Pl}}$. The ratio is $(m_{\text{T}}/M_{\text{Pl}})^2$.

E. Amplitude-frequency tradeoff

The peak amplitude satisfies

$$\Omega_{\text{GW}}^{\text{peak}} \propto f_b^4, \quad (36)$$

where $f_b \propto m_{\text{T}}^{1/2}$. Lowering f_b by a factor of 10 reduces the amplitude by 10^4 . Moving from $f_{\text{Pl}} \sim 10^{10}$ Hz to LISA frequencies (10^{-3} Hz) requires 10^{13} in frequency, giving 10^{52} in amplitude suppression. This tradeoff is universal to *all* vacuum-amplification bounces.

F. Minimum detector gap

The best case is the Einstein Telescope (ET) at $\sim 10^4$ Hz, with optimal $m_{\text{T}} \sim 10^7$ GeV giving $\Omega_{\text{GW}}^{\text{peak}} h^2 \sim 3 \times 10^{-30}$ against a sensitivity of $\sim 10^{-13}$. The gap is 10^{17} —insurmountable.

G. Comparison with other GW sources

The bounce spectrum (flat + exponential cutoff) differs qualitatively from inflation (near-flat power law), phase transitions (peaked with f^3 rise and f^{-1} decay), and cosmic strings (flat without cutoff). However, these shape differences are moot: the amplitude is $\geq 10^{17}$ below any detector threshold. Furthermore, the amplitude scaling $\Omega_{\text{GW}} \propto (H_{\text{bounce}}/M_{\text{Pl}})^2$ applies equally to LQC, ekpyrotic, and any other bounce model relying on vacuum amplification.

H. Barrier 12: Vacuum amplification ceiling

The GW energy density from vacuum amplification at a cosmological bounce satisfies

$$\Omega_{\text{GW}} \propto \left(\frac{H_{\text{bounce}}}{M_{\text{Pl}}} \right)^2 \propto \left(\frac{f_b}{f_{\text{Pl}}} \right)^4. \quad (37)$$

This fundamental bound ensures that any bounce whose features fall in a GW detector band ($f < 10^4$ Hz) produces a signal at least 10^{17} below sensitivity. The bound is model-independent and applies to all bounce mechanisms relying on vacuum amplification.

The only escape would be a *non-vacuum* GW source: a pre-bounce contraction phase that coherently amplifies tensor modes, post-bounce phase transitions, or topological defect formation. All of these either require additional physics beyond the bounce or are not bounce-specific.

XIII. BARYOGENESIS AND RELIC PRODUCTION

The spin-torsion bounce couples directly to fermion chirality through the $(J^5)^2$ contact interaction. At the bounce energy scale ($T \sim M_{\text{Pl}}$ for ECH, $T \sim m_{\text{T}}$ for PGT), particle production and asymmetry generation are kinematically open. This motivates testing whether the bounce can drive baryogenesis, dark matter production, or other relic mechanisms.

A. Candidate mechanisms screened

Seven candidate mechanisms were constructed and screened:

- A. *Axial chemical potential baryogenesis*: The torsion $(J^5)^2$ interaction generates an effective chemical potential $\mu_5 \sim G\xi n_5$ for chiral fermions, biasing sphaleron processes.
- B. *Modified heavy fermion decay*: Bounce-era curvature shifts heavy particle masses by $\delta m \sim \mathcal{O}(M_{\text{Pl}})$, modifying CP-violating decay rates.
- C. *Gravitational superheavy dark matter*: Nonadiabatic particle production at the bounce creates superheavy relics with $m \sim M_{\text{Pl}}$.
- D. *Bounce-modified axion abundance*: The bounce modifies the axion misalignment angle θ_i .
- E. *Torsion-assisted leptogenesis*: The $(J^5)^2$ interaction enhances lepton-number-violating processes.
- F. *PBH formation*: Bounce-scale perturbations collapse to primordial black holes.
- G. *Sterile neutrino production*: Spin-density coupling populates sterile neutrinos during the bounce.

B. Results

All seven candidates are killed.

The $(J^5)^2$ interaction conserves both baryon number B and lepton number L . It is a chirality-sensitive *scattering* operator, not a B - or L -violating operator. Any baryogenesis mechanism must rely on external B/L -violating physics (sphalerons, Majorana masses), with torsion providing at most a perturbative correction.

At $T \sim M_{\text{Pl}}$, all gravitational-strength interactions have comparable rates: $\Gamma \sim G^2 T^5 \sim H$. The torsion $(J^5)^2$ interaction is one of approximately $g_* \sim 100$ such channels, contributing $\sim 1\%$ of any effect. It cannot dominate any relic abundance.

The strongest candidate (A: axial μ_5 baryogenesis) produces $\eta_B \sim 10^{-9}$ for maximal initial chirality—tantalizingly close to observation. But the initial chirality n_5 is a *free parameter* that the bounce does not set (the gravitational chiral anomaly vanishes: $\tilde{R}R = 0$ on FRW, established in Sec. VII). The weakest candidate (D: axion) changes the misalignment angle by $\Delta\theta \sim 10^{-96}$ rad—96 orders of magnitude below relevance.

C. Barrier 13a: Gravitational democracy

At $T \sim M_{\text{Pl}}$, all gravitational-strength interactions have comparable rates. The torsion $(J^5)^2$ interaction contributes $\sim 1/g_$ of the total gravitational effect. It conserves B and L , so it cannot generate asymmetries that generic gravitational interactions do not. No relic abundance or asymmetry is dominated by, or specific to, the torsion interaction.*

The bounce is trapped: too high-energy for late-time observables (scale separation, Branches A–M) and too generic at its own energy scale for early-time observables (gravitational democracy, Branch N).

XIV. HIDDEN-SECTOR VACUUM SELECTION

Branch J established that purely Hamiltonian (reversible) state selection is killed by Liouville’s theorem (Barrier 9). The remaining logical possibility is *irreversible* mechanisms: thermalization, particle production, tunneling, dissipation, or phase-transition hysteresis. We test whether the bounce can trigger such processes in a hidden sector, indirectly selecting the late-time vacuum energy.

A. Candidate mechanisms screened

Seven candidates were constructed:

- A. *First-order phase transition*: Bounce curvature ($R \sim M_{\text{Pl}}^2$) triggers a hidden-sector transition.

- B. *Metastable vacuum trapping*: Bounce temporarily deforms the hidden-sector potential, trapping the field.
- C. *Symmetry restoration/re-breaking*: The bounce restores a hidden symmetry that re-breaks with biased branch selection.
- D. *Tunneling enhancement*: Bounce-era potential distortion enhances or suppresses quantum tunneling.
- E. *Nonadiabatic production*: Bounce populates a pNGB hidden sector via particle production.
- F. *Dissipative reheating*: Post-bounce thermalization selects a thermal branch.
- G. *Domain-wall percolation*: Bounce-imposed asymmetry biases multi-vacuum domain percolation.

B. Results

All seven candidates are killed.

The results partition into three structural outcomes:

(i) *Irreversibility is real but not bounce-specific*. Five candidates (B, C, E, F, G) involve genuine irreversibility, but the same processes occur in any hot cosmology. The bounce provides energy, but so does inflationary reheating.

(ii) *Bounce-specific candidates fail on naturalness*. Two candidates (A, D) have marginal bounce-specificity: only the Planck-scale curvature of the ECH bounce can trigger the transition. But the resulting vacuum energy ρ_{DE} is set by hidden-sector potential parameters, not by the bounce. The bounce determines *whether* a transition occurs, not *what* vacuum energy results.

(iii) *The 10^{122} hierarchy is untouched*. No mechanism connects M_{Pl}^4 to $(2.3 \text{ meV})^4$ through bounce dynamics. The hierarchy must be bridged by hidden-sector parameters that the bounce does not set.

C. Barrier 13b: Bounce–vacuum energy decoupling

The bounce operates at scale M ($= M_{\text{Pl}}$ or m_{T}). The late-time vacuum energy is at scale Λ_{DE}^4 . Any formula $\rho_{\text{DE}} = f(M, g_i)$ requires a suppression factor $\sim (\Lambda_{\text{DE}}/M)^p$ that cannot be generated by bounce dynamics. Adding irreversibility determines whether a transition occurs but not the vacuum energy scale. The trigger and the outcome are structurally decoupled.

D. Completeness of the state-change closure

Barriers 9 and 13 together close *all* state-change mechanisms:

- Reversible (Hamiltonian) mechanisms: killed by Liouville (Barrier 9).
 - Irreversible, bounce-specific mechanisms: killed by decoupling (Barrier 13).
 - Irreversible, non-bounce-specific mechanisms: not relevant to the bounce program.
- There is no remaining category. The logical space of bounce→DE state-change connections is exhausted.

XV. PGT TORSION RELIC COSMOLOGY

The last surviving positive channel was torsion relic cosmology: if the PGT Sector II pseudoscalar torsion mode $\phi(t) \equiv S_0(t)$ carries $\mathcal{O}(1)$ of the bounce energy, its post-bounce decay yields a massive relic constrained by BBN, CMB spectral distortions, and dark radiation bounds. The gating question is whether ϕ is actually populated at the bounce.

A. FRW parity protection

The Sector II pseudoscalar mode $S_0(t)$ is kinematically allowed on FRW: isotropy permits $S_\mu = (S_0(t), 0, 0, 0)$. However, S_0 is parity-*odd* (a pseudoscalar under spatial inversion). FRW spacetime is parity-invariant, and the PGT Sector II action is quadratic in S_μ , possessing an exact $\mathbb{Z}_2$ symmetry $\phi \rightarrow -\phi$. The field equation on FRW is

$$\ddot{\phi} + 3H\dot{\phi} + m_{\text{T}}^2\phi = 0, \quad (38)$$

with *no source term*: parity-even matter cannot source a parity-odd field. The origin $\phi = 0$ is a stable fixed point ($m_{\text{T}}^2 > 0$), and the bounce—being a parity-even gravitational event—cannot dislodge it.

B. Population mechanisms tested

Five population scenarios were assessed:

1. *Self-consistent bounce*: The modified Friedmann equation works with $\phi = 0$ (the contact-term $(J^5)^2$ alone drives the bounce). The propagating mode is not required.
2. *Quantum vacuum fluctuations*: Produce particle pairs (occupation number $\langle n_k \rangle$), not a homogeneous condensate $\langle \phi \rangle \neq 0$.
3. *Chiral matter sourcing*: A chiral asymmetry $n_5 \neq 0$ in the matter sector could source ϕ through the axial coupling, but n_5 is a free parameter (the gravitational chiral anomaly $\tilde{R}R = 0$ on FRW; see Sec. VII) and the sourced amplitude is $\phi \sim G n_5 \ll M_{\text{Pl}}$.
4. *Parametric resonance*: The bounce is ~ 1 oscillation period long; no time for resonant buildup.

5. *Unmotivated initial condition:* $\phi(0) \sim M_{\text{Pl}}$ as a free parameter. Not a prediction—equivalent to assuming the conclusion.

No mechanism populates the pseudoscalar mode at $\mathcal{O}(M_{\text{Pl}})$ or $\mathcal{O}(m_T)$.

C. Literature context

The PGT cosmology literature (Shie, Nester, Yo 2008; Chen *et al.* 2009) uses the parity-*even* trace vector $T_0(t)$, which is naturally nonzero on FRW. This is a fundamentally different mode (Sector I/III), which has ghost issues at the target mass scale. The specific question of pseudoscalar population at a PGT bounce had not been addressed; our answer is negative.

D. Barrier 14: Parity protection of the torsion vacuum

The PGT Sector II pseudoscalar torsion mode has an exact $\mathbb{Z}_2$ parity symmetry $\phi \rightarrow -\phi$ that protects $\phi = 0$ as the unique parity-invariant vacuum on FRW. No parity-even dynamics (including the bounce) can excite the mode. The torsion relic abundance is zero unless parity is explicitly broken by the initial conditions.

This closes the last positive observable channel of the minimal PGT bounce program.

XVI. COMPLETE BARRIER CATALOG

Fourteen structural barriers have been identified across the full research program (Branches A–P). They span five distinct failure modes: energy-budget failure (Barriers 1–7), signature failure (Barriers 8, 10–12), reversible state-selection failure (Barrier 9), irreversible state-selection / relic failure (Barrier 13), and relic population failure (Barrier 14).

XVII. DISCUSSION

A. The generic-specific dichotomy

The central finding of the bounce characterization is a sharp dichotomy: *what is detectable is generic to any bounce; what is specific to torsion is undetectable.*

Observable quantities—the flat tensor spectrum ($n_T \approx 0$), the unit transfer function ($T(k) = 1$), the GW spectral shape (flat + exponential cutoff)—are all shared by any time-symmetric radiation bounce. They are properties of the bounce *geometry*, not the bounce *mechanism*.

Torsion-specific quantities—the critical density $\rho_{\text{crit}} \approx 0.21 M_{\text{Pl}}^4$, the exact bounce profile, the $(J^5)^2$ corrections,

the chirality $\Delta\chi = 0$ —are all either unobservable (confined to GHz scales) or null (parity-even interaction, Pontryagin density vanishes on FRW).

B. Comparison with other bouncing cosmologies

1. Loop quantum cosmology (LQC)

LQC also produces a modified Friedmann equation of the form (4) with $\rho_{\text{crit}} \approx 0.41 \rho_{\text{Pl}}$ (slightly different numerical coefficient). The tensor spectrum has the same functional form: flat, with $P_T \sim 10^{-64}$ and $n_T \approx 0$. The GW amplitude-frequency tradeoff (37) applies identically. The spin-torsion bounce and LQC bounce are observationally indistinguishable at any accessible frequency.

2. Ekpyrotic bounce

Ekpyrotic scenarios produce a strongly blue-tilted tensor spectrum ($n_T \sim 2$), in contrast to the flat spectrum of the spin-torsion and LQC bounces. However, the ekpyrotic bounce amplitude is similarly suppressed. In principle, n_T distinguishes ekpyrotic from radiation-symmetric bounces—but only if the amplitude is detectable, which it is not.

3. Matter bounce

The matter bounce scenario [30, 31] also produces $n_T \approx 0$ (scale-invariant tensors), making it degenerate with the spin-torsion bounce. The matter bounce requires a specific pre-bounce contraction phase (dust-like), which is not part of the minimal spin-torsion model. Both models face the same amplitude suppression.

C. The complementary trap

The spin-torsion bounce is trapped between three complementary failures:

- *Too high-energy* for late-time observables (scale separation, Planck suppression—Branches A–M).
- *Too generic* at its own energy scale (gravitational democracy—Branch N). At $T \sim M_{\text{Pl}}$, torsion is one gravitational-strength channel among ~ 100 .
- *Too decoupled* from late-time vacuum energy (bounce–vacuum decoupling—Branch O). The bounce trigger and the vacuum energy outcome are structurally independent.

TABLE VII. Complete catalog of fourteen structural barriers identified in the spin-torsion bounce and dark energy program.

#	Barrier	Branch	Description
1	Mass-coupling lock	Found. A (PGT)	m_B and g_{eff} decrease together; $m_B/g_{\text{eff}} \sim M_{\text{Pl}}^2$ fixed
2	Topological-shift duality	Found. B	Nieh-Yan non-topological in MAG but geometric content blocked
3	Scalar-tensor universality	Found. C	Environmental mass reduces to standard scalar-tensor on FRW
4	Planck suppression	Found. D	Connection coupling gives 1 $\partial\phi/\text{vertex}$; disformal requires 2
5	Scale separation	Found. E	$k_{\text{CMB}}/k_b \sim 10^{-28}$; global integrals insensitive to bounce
6	Attractor-sensitivity dilemma	Found. F	Attractors erase IC; sensitivity requires fine-tuning
7	Parameter immunity	Found. G	Cyclic incompatibility; continuous vacuum solutions
8	Parity-even effective interaction	Branch H	$(J^5)^2$ is parity-even; FRW kills spatial P-odd backgrounds
9	Hamiltonian phase-space conservation	Branch J	Liouville prevents phase-space contraction at bounce
10	UV $\rightarrow$ IR specificity dilemma	Branch L ₁	Observable $\Rightarrow$ generic; specific $\Rightarrow$ non-minimal
11	Decoupling universality	Branch L ₂	Light PGT gauge fields decouple universally
12	Vacuum amplification ceiling	Branch M	$\Omega_{\text{GW}} \propto (H/M_{\text{Pl}})^2$; minimum 10^{17} gap to detectors
13	Gravitational democracy / bounce-vacuum decoupling	Branches N, O	At $T \sim M_{\text{Pl}}$, torsion is 1 of ~ 100 channels; bounce trigger decoupled from vacuum energy outcome
14	Parity protection of torsion vacuum	Branch P	$\mathbb{Z}_2$: $\phi \rightarrow -\phi$ protects $\phi = 0$; no population mechanism

D. What would change the assessment

The assessment would change if:

1. A *non-vacuum* GW source is identified at the bounce (particle production, phase transition) that does not suffer from the vacuum amplification ceiling. This would require a detailed model of the bounce-to-inflation transition.
2. A *pre-bounce* contraction phase with $w \neq 1/3$ is specified, allowing horizon exit of tensor modes at CMB scales during contraction. This is additional physics beyond the spin-torsion bounce.
3. A mechanism breaking the mass-coupling lock is found (independent mass-generating mechanism), enabling observable torsion-specific signatures.
4. Non-perturbative or multi-loop effects generate γ -dependent vacuum structure not accessible at the tested orders.
5. A mechanism is found that connects the bounce energy scale to the vacuum energy scale without fine-tuning, evading both Liouville (Barrier 9) and bounce-vacuum decoupling (Barrier 13).

None of these outcomes is assumed or expected within the minimal and near-minimal program assessed here.

E. The program as a publishable negative result

Negative results in theoretical physics are undervalued but scientifically important. The combined program (four minimal DE routes, seven extended foundations, five bounce observable channels, seven baryogenesis/relic candidates, seven hidden-sector vacuum candidates) establishes:

- A complete classification of failure modes (energy-budget, signature, reversible state-selection, irreversible state-selection / relic production).
- Fourteen structural barriers spanning five failure modes.
- Six structural lessons and five decision requirements (DR1–DR5) constraining future proposals.
- A reproducible methodology for disciplined model testing.
- Exhaustive coverage of the logical space: direct derivation (A–G), observable signatures (H, K, L–M), state selection (J, O), UV $\rightarrow$ IR bridge (L), relic production (N), and GW spectrum (M).

The program's value lies not in any single null result but in the *completeness* of the closure: every natural direction has been tested, and the structural reasons for failure have been identified and cataloged.

XVIII. CONCLUSION

This paper presents a comprehensive theoretical assessment of the Einstein-Cartan-Holst framework as both a dark-energy phenomenology and a bouncing cosmology. The contribution is threefold.

First, the phenomenological dark-energy ansatz rooted in ECH gravity remains observationally consistent. It reduces fine-tuning from 10^{120} to $\sim 10^5$, yields MCMC fits partially reducing the H_0 and σ_8/S_8 tensions, and has a parity-odd structure qualitatively consistent with observed birefringence. The equation of state $w = -1$ is assumed, not derived.

Second, the systematic closure of four minimal and seven extended derivation routes establishes that neither the minimal ECH model nor its most natural extensions can derive late-time dark energy from geometry. Seven

structural barriers and six structural lessons map the failure landscape. The mass-coupling lock (Barrier 1)—in which a light torsion mass automatically decouples the field from matter—is a previously unrecognized obstruction that applies to any geometric theory where mass and coupling share a common kinetic-normalization origin.

Third, the comprehensive characterization of the spin-torsion bounce across all observable channels establishes its observational inertness. The tensor spectrum is $P_T \approx 2 \times 10^{-64}$ (undetectable). The chirality is $\Delta\chi = 0$ exactly (no parity violation). The scalar transfer function is $T(k) = 1$ (no bounce imprint). State selection is prevented by Liouville’s theorem. The PGT gravitational wave spectrum has a minimum 10^{17} gap to any detector. Baryogenesis and relic production via the $(J^5)^2$ interaction are killed by gravitational democracy at $T \sim M_{\text{Pl}}$. Hidden-sector vacuum selection via irreversible transitions is killed by bounce–vacuum energy decoupling: the bounce determines *whether* a transition occurs, not *what* vacuum energy results. The last positive channel—PGT torsion relic cosmology—is closed by $\mathbb{Z}_2$ parity protection: the pseudoscalar mode cannot be populated at the bounce without explicit parity violation in the initial conditions. Seven additional barriers (8–14) are cataloged.

The spin-torsion bounce is theoretically consistent but observationally inert. Fourteen structural barriers, spanning ten research branches and five failure modes, prevent any nontrivial connection between the Planck-scale bounce and low-energy cosmological observations. The logical space of bounce→observable connections is exhausted: direct derivation, observable signatures, state selection (both reversible and irreversible), relic production, torsion relic population, and UV→IR bridges have all been tested and closed. Future directions require physics beyond minimal Einstein-Cartan gravity and quadratic Poincaré gauge theory.

ACKNOWLEDGMENTS

The author thanks the anonymous referees for constructive feedback. Computational resources for MCMC analyses were provided by RunPod. This work made use of Cobaya [21] and CAMB.

Appendix A: Notation and Conventions

Natural units $c = \hbar = 1$ throughout, with $8\pi G = M_{\text{Pl}}^{-2}$. Metric signature $(-, +, +, +)$. Reduced Planck mass $M_{\text{Pl}} = (8\pi G)^{-1/2} = 2.435 \times 10^{18}$ GeV. $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ with $(\gamma^5)^2 = 1$. The vacuum energy density is $\rho_\Lambda = \Lambda_{\text{eff}} M_{\text{Pl}}^2/(8\pi)$; when we write $\rho_\Lambda \approx (2.3 \text{ meV})^4$ this refers to the energy density. Greek indices $\mu, \nu, \dots$ denote spacetime coordinates; Latin indices $I, J, \dots$ denote tangent-space (Lorentz) indices. The Levi-Civita symbol ε^{IJKL} is totally antisymmetric with $\varepsilon^{0123} = +1$.

Key symbols:

- γ —Barbero-Immirzi parameter (0.274 ± 0.020)
- K^{IJ}_μ —contorsion tensor
- T^I_{JK} —torsion tensor
- α/M —parity-odd coefficient (mass dimension -1)
- $\mathcal{D}_{\text{inf}}$ —inflationary dilution factor ($\sim e^{-3N_{\text{tot}}}$)
- $\Xi \equiv [(\alpha/M) M_{\text{Pl}}] \mathcal{D}_{\text{inf}}$ —inflationary suppression factor; $\rho_\Lambda = \Xi M_{\text{Pl}}^4$
- $G_{\text{eff}} = (3\kappa^2/16) \gamma^2/(\gamma^2 + 1)$ —effective four-fermion coupling
- $G_{\text{SP}} \propto (\gamma^2 - 1)/(\gamma^2 + 1)$ —scalar/pseudoscalar channel coupling
- m_T —PGT torsion mass
- g_{eff} —canonically normalized matter coupling
- ρ_{crit} —bounce critical density
- k_b —bounce characteristic wavenumber
- f_b —bounce characteristic frequency

Abbreviations: LQG (Loop Quantum Gravity), EC (Einstein-Cartan), ECH (Einstein-Cartan-Holst), PGT (Poincaré Gauge Theory), ALP (axion-like particle), NJL (Nambu-Jona-Lasinio), MAG (metric-affine gravity), MCMC (Markov Chain Monte Carlo), ET (Einstein Telescope), HF (high-frequency).

Appendix B: Parameter Summary

Appendix C: Claims Classification

Appendix D: Bardeen Equation Through the Bounce

We sketch the derivation of the Bardeen equation on the spin-torsion bounce background and the proof of $T(k) = 1$.

1. Setup

Consider scalar perturbations of a radiation fluid ($w = 1/3$, $c_s^2 = 1/3$) on the modified Friedmann background (4). In the longitudinal gauge ($B = E = 0$), the perturbed metric is

$$ds^2 = -(1 + 2\Phi) dt^2 + a^2(1 - 2\Psi) \delta_{ij} dx^i dx^j, \quad (\text{D1})$$

with $\Phi = \Psi$ for a perfect fluid (no anisotropic stress).

TABLE VIII. Complete parameter classification. Status categories: *Fixed* (external input), *Ansatz* (phenomenological scaling), *Assumed* (not derived), *Fit* (MCMC), *Derived* (from other parameters).

Parameter	Description	Value	Status	Notes
<i>Framework parameters</i>				
γ	Barbero-Immirzi	0.274 ± 0.020	Fixed	LQG area spectrum
α/M	Parity-odd coeff.	$\sim 10^{-33} \text{ eV}^{-1}$	Fit	One-loop motivated
N_{tot}	Total e -folds	~ 92	Assumed	Fitted to ρ_Λ
$\mathcal{D}_{\text{inf}}$	Inflation dilution	$\sim e^{-3N_{\text{tot}}}$	Derived	From N_{tot}
Ξ	Infl. suppression	$\sim 10^{-123}$	Ansatz	$= [(\alpha/M)M_{\text{Pl}}]\mathcal{D}_{\text{inf}}$
w	Equation of state	-1	Assumed	Not derived
<i>Standard cosmological parameters (MCMC)</i>				
H_0	Hubble constant	$69.2 \pm 0.8^{\text{a}}$	Fit	km/s/Mpc
σ_8	Clustering ampl.	$0.785 \pm 0.016^{\text{a}}$	Fit	
S_8	$\sigma_8(\Omega_m/0.3)^{0.5}$	0.80 ± 0.02	Derived	
Ω_m	Matter density	0.310 ± 0.008	Fit	
$\Omega_b h^2$	Baryon density	0.02237 ± 0.00015	Fit	
n_s	Spectral index	0.9649 ± 0.0042	Fit	
τ	Optical depth	0.054 ± 0.007	Fit	
<i>Extended/bounce parameters</i>				
ΔN_{eff}	Extra species	$\sim 0^{\text{a}}$	Fit	Consistent with 0
ρ_{crit}	Bounce density	$0.21 M_{\text{Pl}}^4$	Derived	Minimal ECH
m_{T}	PGT torsion mass	Free	Model param.	Sets PGT bounce scale
$ t_3 $	PGT coupling	Free	Model param.	$m_{\text{T}} = M_{\text{Pl}}/(2\sqrt{ t_3 })$

^aFull tension dataset (includes SH0ES H_0 prior). Verification finds $H_0 = 67.68 \pm 1.06$, $\Delta N_{\text{eff}} = -0.020 \pm 0.169$.

2. Derivation of the Bardeen equation

From the linearized Einstein equations on the modified Friedmann background, one obtains [28, 29]:

$$\ddot{\Phi} + 5H\dot{\Phi} + \left[\frac{k^2}{3a^2} + 2\dot{H} + 4H^2 \right] \Phi = 0. \quad (\text{D2})$$

The key feature for bouncing cosmologies: all coefficients are *regular* at $H = 0$. Specifically:

- The friction term $5H\dot{\Phi}$ vanishes at $H = 0$ (the bounce instant).
- The effective mass term $2\dot{H} + 4H^2$ remains finite: $\dot{H}(0) \approx 3.52 M_{\text{Pl}}^2$ and $H(0) = 0$.
- The gradient term $k^2/(3a^2)$ is regular and positive.

This regularity is in contrast to the Mukhanov-Sasaki variable $v = z\dot{\mathcal{R}}$ where $z = a\dot{\phi}/H \rightarrow \infty$ at the bounce.

3. Time-reversal symmetry and $T(k) = 1$

The bounce background $a(t)$ is an even function of t : $a(-t) = a(t)$. Therefore $H(-t) = -H(t)$ and $\dot{H}(-t) = \dot{H}(t)$. Under the substitution $t \rightarrow -t$, Eq. (D2) maps to

$$\ddot{\Phi} - 5H\dot{\Phi} + \left[\frac{k^2}{3a^2} + 2\dot{H} + 4H^2 \right] \Phi = 0, \quad (\text{D3})$$

where the only change is the sign of the friction term. This means that if $\Phi_{\text{grow}}(t)$ is a growing solution in the

contracting phase ($t < 0$, $H < 0$, friction term is anti-damping), then $\Phi_{\text{grow}}(-t)$ is a *decaying* solution in the expanding phase ($t > 0$).

The constant mode Φ_{const} (the physically relevant mode for the power spectrum) is symmetric under $t \rightarrow -t$ and maps to itself with unit amplitude. Therefore:

$$T(k) \equiv \frac{\Phi_{\text{const}}^{\text{out}}}{\Phi_{\text{const}}^{\text{in}}} = 1 \quad \text{for all } k \ll k_b. \quad (\text{D4})$$

Appendix E: PGT Ghost-Free Conditions

1. Quadratic PGT action

The most general quadratic PGT action with torsion-squared and curvature-squared terms has ten free parameters (three torsion-squared a_1, a_2, a_3 plus six curvature-squared b_i plus Λ_0) [23, 25, 26].

2. Single-mode ghost-free models

Sezgin and van Nieuwenhuizen [23] identified the conditions for ghost-freedom (absence of negative-norm states) in the linearized theory around flat spacetime. Karananas [24] confirmed and extended this analysis. The cleanest single-mode models are:

- *Spin-0⁻ (axial pseudoscalar)*: Ghost-free for all $|t_3| > 0$. Mass $m_B = M_{\text{Pl}}/(4\sqrt{\pi|t_3|})$. This is the mode assessed in Foundation A.

TABLE IX. Expanded claims classification. Categories: Derived (structural results), Phenomenological (ansatz/fit/assumption), Negative (clean null results from the testing program), and Retired (superseded by closures).

Claim	Status	Basis	Notes
<i>Derived (structural results)</i>			
ECH action structure	Derived	Standard EC + Holst	Background
Torsion is algebraic (minimal ECH)	Derived	EC field equations	Structural
Perfect-square four-fermion	Derived	Torsion elimination	Exact result
$G_{\text{SP}} < 0$ at $\gamma < 1$	Derived	Fierz rearrangement	Track B closure
One-loop $\mathcal{D}$ is γ -free	Derived	Torsion elimination	G v1 closure
Dyn. Immirzi = ALP	Derived (lit.)	Calcagni-Mercuri 2009	T1 closure
No photon coupling in min. ECH	Derived	Action structure	S1 closure
Nieh-Yan is topological in 4D	Derived (lit.)	Differential geometry	T1 closure
$(J^5)^2$ is parity-even	Derived	Algebra	Branch H closure
$T(k) = 1$ (time-reversal)	Derived	Analytic + numerical	Branch K
$\Delta\chi = 0$ on FRW	Derived	FRW isotropy	Branch H closure
Liouville prevents state selection	Derived	Hamiltonian mechanics	Branch J closure
$\Omega_{\text{GW}} \propto (m_{\text{T}}/M_{\text{Pl}})^2$	Derived	Bogoliubov calculation	Branch M
<i>Phenomenological (ansatz, fit, or assumption)</i>			
$\rho_{\Lambda} = \Xi M_{\text{Pl}}^4$	Ansatz	Scaling argument	Not a derivation
$w = -1$ at late times	Assumed	Observational input	Not derived
H_0, σ_8, S_8 values	Fit	MCMC	Standard analysis
Fine-tuning $10^{120} \rightarrow 10^5$	Parametric	Sensitivity scan	Valid
<i>Negative results (clean null)</i>			
$P_T \approx 2 \times 10^{-64}$	Null (amplitude)	Numerical	Undetectable
$n_T \approx 0$ (generic)	Null (shape)	Numerical + analytic	Any rad. bounce
$r \sim 10^{-55}$	Null (ratio)	Derived	Undetectable
$f_{\text{NL}} \sim 10^{-56}$	Null (NG)	Order-of-magnitude	Undetectable
GW gap $\geq 10^{17}$	Null (GW)	Bogoliubov + detector	Branch M
No Horndeski constraint	Null (compat.)	Scale separation	Branch I
<i>Retired (closed by testing program)</i>			
Birefringence as prediction	Retired	S1 closure	No photon coupling
γ drives late-time physics	Retired	B, G, T1 closures	UV-scale only
Condensate mechanism	Retired	Track B closure	$G_{\text{SP}} < 0$
PGT torsion as DE mechanism	Retired	Foundation A	Mass-coupling lock
Bounce tensor as observable	Retired	Branch H	$P_T \sim 10^{-64}$
Bounce chirality as observable	Retired	Branch H	$\Delta\chi = 0$
Bounce state selection	Retired	Branch J	Liouville theorem
PGT GW as observable	Retired	Branch M	Vac. ampl. ceiling

- *Spin-0⁺ (scalar)*: Ghost-free only in a restricted parameter range. Develops a ghost at the mass scale needed for observable bounce features. Rejected.
- *Spin-2⁺ (tensor)*: Ghost-free but massive spin-2 fields face the van Dam-Veltman-Zakharov discontinuity and require careful Vainshtein screening.

3. Sector II analysis

For the spin-0⁻ mode in Sector II, the propagator is

$$D(p^2) = \frac{1}{p^2 - m_B^2}, \quad (\text{E1})$$

with no ghost pole (residue is positive). The canonically normalized field $\phi_{\text{can}} = \sqrt{Z} \phi$ with $Z \propto |t_3|$ gives

$$m_B = \frac{M_{\text{Pl}}}{4\sqrt{\pi}|t_3|}, \quad g_{\text{eff}} = \frac{g}{\sqrt{Z}} \sim \frac{1}{M_{\text{Pl}}\sqrt{|t_3|}}, \quad (\text{E2})$$

confirming the mass-coupling lock (12).

4. Symmetry analysis

Five candidate symmetries for protecting m_B were investigated:

1. *Gauge symmetry*: The 0⁻ mode is not a gauge field; no gauge invariance protects its mass.
2. *Shift symmetry*: The mass term $m_B^2 \phi^2$ explicitly breaks any shift symmetry $\phi \rightarrow \phi + c$.
3. *Chiral symmetry*: The axial current coupling preserves chiral symmetry, but this does not protect a scalar mass.
4. *Topological protection*: No topological mechanism applies in 4D for a massive pseudoscalar.

5. *Supersymmetry*: Would pair the 0^- with a fermion, but SUSY is not part of PGT and its breaking scale is unknown.

The 't Hooft naturalness criterion fails: setting $m_B = 0$ does not restore any symmetry of the PGT action. Gravi-

ton loops generate

$$\delta m_B^2 \sim \frac{M_{\text{Pl}}^2}{16\pi^2}, \quad (\text{E3})$$

requiring fine-tuning of order 10^{-57} to maintain $m_B \sim \text{meV}$.

-
- [1] S. Weinberg, “The cosmological constant problem,” *Rev. Mod. Phys.* **61**, 1 (1989).
- [2] Planck Collaboration (N. Aghanim *et al.*), “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [3] A. G. Riess *et al.*, “A comprehensive measurement of the local value of the Hubble constant with 1 km/s/Mpc uncertainty from the Hubble Space Telescope and the SH0ES team,” *Astrophys. J. Lett.* **934**, L7 (2022).
- [4] KiDS Collaboration (M. Asgari *et al.*), “KiDS-1000 cosmology: Cosmic shear constraints on the amplitude of matter fluctuations,” *Astron. Astrophys.* **645**, A104 (2021).
- [5] DES Collaboration (T. M. C. Abbott *et al.*), “Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing,” *Phys. Rev. D* **105**, 023520 (2022).
- [6] DESI Collaboration, “DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations,” *arXiv:2404.03002* (2024).
- [7] DESI Collaboration, “DESI DR2 results I: Baryon acoustic oscillations and cosmological constraints,” *arXiv:2503.14738* (2025).
- [8] A. Ashtekar and P. Singh, “Loop quantum cosmology: A status report,” *Class. Quant. Grav.* **28**, 213001 (2011).
- [9] L. Freidel, D. Minic, and T. Takeuchi, “Quantum gravity, torsion, parity violation and all that,” *Phys. Rev. D* **72**, 104002 (2005).
- [10] R. Mercuri, “Fermion interactions, cosmological constant and torsion,” *Phys. Rev. D* **73**, 084016 (2006).
- [11] R. Mercuri, “Peccei-Quinn mechanism in gravity and the nature of the Barbero-Immirzi parameter,” *Phys. Rev. Lett.* **103**, 081302 (2009).
- [12] N. J. Popławski, “Cosmological constant from quarks and torsion,” *Phys. Rev. D* **85**, 107502 (2012).
- [13] I. L. Shapiro and P. M. Teixeira, “Quantum Einstein-Cartan theory with the Holst term,” *Class. Quant. Grav.* **31**, 185002 (2014).
- [14] I. L. Shapiro, “Physical aspects of the space-time torsion,” *Phys. Rept.* **357**, 113 (2002).
- [15] D. V. Vassilevich, “Heat kernel expansion: User’s manual,” *Phys. Rept.* **388**, 279 (2003).
- [16] G. Calcagni and R. Mercuri, “Barbero-Immirzi field in canonical formalism of pure gravity,” *Phys. Rev. D* **79**, 104003 (2009).
- [17] V. Taveras and N. Yunes, “Barbero-Immirzi parameter as a scalar field: K-inflation from loop quantum gravity?” *Phys. Rev. D* **78**, 064070 (2008).
- [18] J. R. Eskilt and E. Komatsu, “Improved constraints on cosmic birefringence from the WMAP and Planck cosmic microwave background polarization data,” *Phys. Rev. D* **106**, 063503 (2022).
- [19] P. Diego-Palazuelos and E. Komatsu, “Cosmic birefringence from ACT DR6,” (2025).
- [20] H. Golden, “Geometric Dark Energy from Spin-Torsion Cosmology: Phenomenological Constraints and Correlated Signatures,” Preprint HUBIFY-2026-001, v1.6 (2026).
- [21] J. Torrado and A. Lewis, “Cobaya: Code for Bayesian analysis of hierarchical physical models,” *JCAP* **05**, 057 (2021).
- [22] J. Liu, R. Niu, and R.-G. Cai, “Torsion cosmology with DESI DR2,” *arXiv:2503.02599* (2025).
- [23] E. Sezgin and P. van Nieuwenhuizen, “New ghost-free gravity Lagrangians with propagating torsion,” *Phys. Rev. D* **21**, 3269 (1980).
- [24] G. K. Karananas, “The particle spectrum of parity-violating Poincaré gravitational theory,” *Class. Quant. Grav.* **32**, 055012 (2015).
- [25] K. Hayashi and T. Shirafuji, “Gravity from Poincaré gauge theory of the fundamental particles. I. General formulation,” *Prog. Theor. Phys.* **64**, 866 (1980).
- [26] D. E. Neville, “Gravity theories with propagating torsion,” *Phys. Rev. D* **21**, 867 (1980).
- [27] H.-J. Yo and J. M. Nester, “Hamiltonian analysis of Poincaré gauge theory,” *Int. J. Mod. Phys. D* **8**, 459 (1999).
- [28] V. F. Mukhanov, H. A. Feldman, and R. H. Brandenberger, “Theory of cosmological perturbations,” *Phys. Rept.* **215**, 203 (1992).
- [29] P. Peter and N. Pinto-Neto, “Primordial perturbations in a non-singular bouncing universe model,” *Phys. Rev. D* **78**, 063506 (2008).
- [30] J. Quintin, Y.-F. Cai, and R. H. Brandenberger, “Matter creation in a nonsingular bouncing cosmology,” *Phys. Rev. D* **90**, 063507 (2015).
- [31] Y.-F. Cai, D. A. Easson, and R. Brandenberger, “Towards a nonsingular bouncing cosmology,” *JCAP* **08**, 020 (2012).
- [32] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, “General relativity with spin and torsion: Foundations and prospects,” *Rev. Mod. Phys.* **48**, 393 (1976).
- [33] N. Kaloper and A. Padilla, “Sequestering the standard model vacuum energy,” *Phys. Rev. Lett.* **112**, 091304 (2014).
- [34] N. J. Popławski, “Universe in a black hole in Einstein-Cartan gravity,” *Astrophys. J.* **832**, 96 (2016).
- [35] LISA Collaboration, “Laser Interferometer Space Antenna,” *arXiv:1702.00786* (2017).
- [36] M. Punturo *et al.*, “The Einstein Telescope: A third-generation gravitational wave observatory,” *Class. Quant. Grav.* **27**, 194002 (2010).
- [37] D. Saadeh *et al.*, “How isotropic is the universe?” *Phys.*

- Rev. Lett. **117**, 131302 (2016).
- [38] C. Legner, W. J. Handley, W. E. V. Barker, and S. S. Ormondroyd, “Torsion condensation in Poincaré gauge theory,” arXiv:2507.09228 (2025).
 - [39] L. Fabbri, “Propagating torsion as an axial-vector field,” arXiv:2502.17979 (2025).
 - [40] F. Izaurieta *et al.*, “Torsion fermion interaction and the Hubble parameter,” Phys. Rev. D **101**, 064007 (2020).
 - [41] S. Akhshabi and E. Zamani, “Cosmological constraints on Einstein-Cartan torsion,” Class. Quant. Grav. **40**, 065006 (2023).